# Siheng Ye

Electrical and Electronic Engineering | Robotics, Control and Intelligent Systems
Ningbo, China | Portfolio | LinkedIn | GitHub | ORCID

## Education

**University of Nottingham Ningbo China** Expected July 2027
BEng (Hons), Electrical and Electronic Engineering | Final-year undergraduate
*Research interests:* Embedded perception, robotic systems and control, cyber-physical systems, and edge AI.
*Selected final-year modules (2026/27, enrolled):* Internet of Things (EEEE3124); Control Systems Design (EEEE3066); Robotics, Dynamics and Control (EEEE3076); Sensing Systems and Signal Processing (EEEE3110).

## Final-Year Project

**PMSM Model-Predictive Control with Digital-Delay Compensation**
*In progress | Supervised by Ahmed Nasr Abdelwahed*
A simulation-based study of digital-delay compensation and prediction robustness in PMSM predictive torque control. Current results are exploratory; hardware validation remains future work.

## Research Experience

**Robot–Elevator Interaction** June–August 2026
*Lift Robot project | Supervised by David Chieng*

- Designed the overall ROS 2 perception-to-action architecture on Jetson AGX Orin; led implementation and testing, including hand-eye calibration, RGB-D localisation, and operator-selected, guarded arm motion.
- The project reported 404/475 correct offline target selections (85.1%), separately from two staged physical button-activation demonstrations. Explored zero-shot detection with YOLO-World.

**Raspberry Pi 5 Engineering Assistant** June–August 2026
*FoSE Undergraduate Research Practice | Supervised by Ian Pratt*

- Led end-to-end implementation and manuscript writing for a four-bit Qwen3-1.7B assistant with course-material retrieval, calculation tools, and a web interface exposing answers, sources, and tool traces.
- Compared seven configurations on 96 EEE questions in an offline GPU-reference evaluation; tool-augmented RAG scored 0.777/1.000. Separate Pi 5 CPU-only, closed-book Base inference scored 0.498/1.000; both are mean rubric scores, not accuracy or web-prototype performance.

**Bangbang V2: Inclusive Teleoperation Robot** March–May 2026
*Supervised by David Chieng; co-supervised by Luke Younghoon Chang*

- Owned the team's vision component: facial and expression recognition using MediaPipe's 478-landmark front end, CNN/ONNX, and interpretable FACS/action-unit routes.
- Implemented and tested the live Pi 5 camera vision path; outputs remained advisory, with final decisions retained by the human operator. FURP 2026 poster (PDF).

**Transformable Land–Air Robot** June–August 2025
*Supervised by Chiew-Foong Kwong; co-supervised by Fuhua Jia*

- Contributed mechanical modifications, cross-modal motion control, and embedded debugging to a FURP 2025 team adaptation of Michael Rechtin's Transforming Quadcopter, combining ground travel and quadrotor flight.

## Selected Engineering Projects

**Closed-Loop Forward Converter** March–May 2026

*EEEE2074 Energy Project | Group 07*

- Designed a 50 W two-switch forward converter and Type III compensation using the Venable K-factor method within Group 07; modelled parasitic circuit elements in PLECS.
- Wound transformer and inductor hardware, characterised magnetic components with an LCR meter, and investigated output regulation and switching-voltage spikes.

**Doppler Radar Speed Detector** October–November 2025

*Embedded signal processing and digital logic project | Group 02*

- Integrated HB100 radar, analogue conditioning, STM32U575 acquisition, CMSIS-DSP FFT processing, and a CPLD display within Group 02.
- Implemented UART reception and display logic in Verilog; investigated signal thresholds and communication timing with an oscilloscope.

**Robotic Vehicle Design** October 2024–March 2025

*EEEE1039 Applied EEE Construction Project | Group 18*

- Contributed PID line-following, OpenCV colour recognition, wireless communication, PWM motor control, and Arduino/Raspberry Pi integration within a four-person team.

## Accepted Conference Contributions

**S. Ye**, Z. Tong, S. Zhang and D. Chieng. "Robot–Elevator Interaction: From Physical Panel Recognition to Activation" (Paper 105). 2026.
*Accepted for oral presentation, IHCI 2026.*

**S. Ye**, X. Li, C. Qi, L. Yin, Y. Zhang and I. Pratt. "A Lightweight Test and Training Platform Utilizing a Raspberry Pi 5: AI-retrieval, Tools and a Compact LLM" (Paper 106). 2026.
*Accepted for oral presentation, IHCI 2026.*

Y. Chang, **S. Ye**, Y. Cao, et al. "Bangbang V2: On-Device Facial Emotion and Bilingual Speech Recognition for a Raspberry Pi-Based Tele-Operated Avatar Robot" (Paper 230). 2026.
*Accepted for presentation, HRFEST 2026.*

Conference acceptance does not establish proceedings publication or indexing.

## Technical Skills

**Programming:** Python, C++, Verilog; MATLAB/Simulink; Linux and SSH.

**Robotics and embedded systems:** ROS 2, Raspberry Pi, Jetson AGX Orin, STM32, Arduino, PID control, RGB-D localisation, and sensor integration.

**Machine learning:** OpenCV, MediaPipe, ONNX, RAG, supervised fine-tuning, GGUF/llama.cpp deployment, and comparative evaluation.

**Electronics and tools:** PLECS, LTspice, Multisim, Git, oscilloscopes, and LCR measurement.

## Languages

**Chinese:** Native or bilingual proficiency.

**English:** Native or bilingual proficiency.

**French:** Professional working proficiency.

**Japanese:** Limited working proficiency.

**Korean:** Limited working proficiency.